

Business Problem Statement

End-to-End Retail / E-commerce Data Analysis

1. Project Positioning

Indian_Retail_Ecommerce_Sales_Customer_Product_Analysis is an **end-to-end data analysis project** focused on Indian retail/e-commerce sales, customer and product performance. The complete analytical workflow is developed in **Google Colab**, starting from raw data ingestion and continuing through data cleaning, validation, exploratory analysis, business analysis, KPI calculation and final insight generation.

The project is not positioned as a Python-only coding exercise. Python is the analytical technology used inside Google Colab; the project itself represents the complete data-analysis lifecycle from raw data to business insights.

2. Business Problem

Retail businesses collect large volumes of transaction data containing customer, product, location, pricing, discount, payment, channel, segment, delivery and rating information. Raw transaction records, however, do not directly explain the major drivers of sales or where the business should focus its attention.

The business needs a structured end-to-end analysis that converts transaction-level data into reliable KPIs, trends, comparisons and actionable findings across sales, customers, products, geography and operations.

3. Business Questions

- Which product categories generate the highest sales?
- Which products generate the highest sales and which are weaker performers?
- Which states and cities contribute the most sales?
- Which payment methods generate the most sales and orders?
- How do Online and Store sales compare?
- Which customer segments contribute the most sales?
- How do sales and orders vary by delivery status?
- How do sales change month by month and year by year?
- Which age groups and genders contribute more sales?
- How does category performance vary by gender?
- Which state-category combinations perform strongly?
- How are discount, rating, quantity and customer age related to sales?
- How do monthly sales vary by category and sales channel?
- How do customer segments perform across sales channels?
- How do delivery statuses vary across sales channels?

4. End-to-End Analytical Objectives

- Ingest and inspect the raw retail dataset in Google Colab.
- Assess data quality before analysis.
- Clean missing values, duplicates, inconsistent categories and formatting issues.
- Validate numeric, categorical and date-related fields.

- Investigate outliers using IQR-based analysis.
- Create useful analytical fields such as Month, Year and Age_Group.
- Perform descriptive, univariate, bivariate and relationship analysis.
- Calculate business KPIs and rankings.
- Identify the strongest patterns across customers, products, categories, regions, channels and operations.
- Export the cleaned dataset for downstream use.

5. Data Quality Outcome

The final quality check in the notebook reports **100,000 rows, 19 original columns, zero missing values, zero duplicate rows and zero duplicate Order IDs**. The cleaned dataset is exported as **cleaned_sales_data.csv**.

6. Business KPIs Produced

Total Sales: **■2,182,312,374.25**

Total Orders: **100,000**

Average Sales per Order: **■21,823.12**

Total Quantity Sold: **180,987**

7. Key Findings Produced

The notebook identifies Electronics as the top category, Odisha as the top state, Cuttack as the top city, Online as the top sales channel, Consumer as the top customer segment, UPI as the top payment method and Delivered as the top delivery status based on the notebook's sales aggregations.

8. Business Value

The project transforms raw retail transactions into a structured view of sales performance, customer behavior, product performance, regional contribution, channel performance and operational patterns. These findings provide a foundation for management reporting, performance monitoring and further decision-making.